

Classification of Motor Imagery EEG Based on a Time-Frequency Analysis and Second-Order Blind Identification

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Abstract—In this paper, two methods for the classification of motor imagery electroencephalogram (EEG) recordings in a brain computer interface (BCI) task was described. The first one is based on a time-frequency analysis of EEG signals. The original EEG signals are converted to time-frequency signals by a function of short time Fourier transforms (STFTs). In another method, we applied second-order blind identification (SOBI), a blind source separation (BSS) algorithm to preprocess EEG data. Subsequently in both of two methods, Fisher class separability criterion was used to select the features. Finally, classification of Motor Imagery EEG evoked by a sequence of randomly mixed left and right motor imagery was performed by a linear classifier or back-propagation neural networks (BPNN), using as inputs the two STFTs time-frequency signals or the two SOBI-recovered SI components or the two EEG channels C3/C4. The results showed that classification accuracy of Motor Imagery EEG was significantly improved by STFTs or SOBI preprocessing.

Keywords—motor imagery EEG; brain computer interface; short time Fourier transforms (STFTs); second-order blind identification; back-propagation neural networks

I. INTRODUCTION

Brain-computer interface (BCI) is an area of research that has recently received much attention. A brain-computer interface is a way of using an individual's thought processes to control computer or electromechanical hardware without using overt muscle activities [1-3]. This type of system has the potential to provide a new form of communication and control options for individuals paralyzed from high-level spinal cord injury, severe neuromuscular disorders, or amyotrophic lateral sclerosis (ALS). The methods and utility of a BCI are currently being investigated in the field of rehabilitation [3-5] and neurorobotics [6-8].

A kind of Analysis brain-computer interface system based on analysis of EEG. Various signal processing methods have been used in EEG analysis for best classification accuracy, such as based on event-related (de-)synchronization (ERD/ERS)[9-11], based on different rhythms of EEG [9,11], based on wavelet transform [12], based on wavelet entropy [13] and have achieved certain results.

II. DATA DESCRIPTION

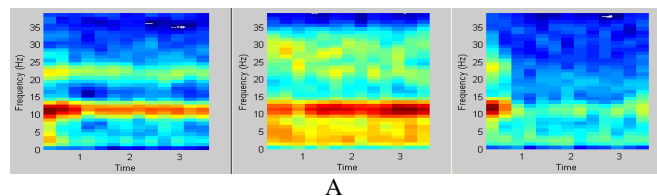
The dataset of BCI competition 2003 is used in this investigation. This dataset was provided by Graz University of Technology. The subject sat in a relaxing chair with armrests. The task was to perform imagery left hand or right hand movements according to a cue. The order of cues was random. The experiment consists of several runs (≥ 6) with 40 trials each after each; after trial begin, the first 2s were quite, at $t=2s$ an acoustic stimulus indicated the beginning of the trial, and a cross "+" is displayed; then from $t=3s$ an arrow to the left or right was displayed for 1s; at the same time the subject was asked to imagine a left hand or right hand, respectively, until the cross disappeared at $t=7s$. Each of the 2 cues was displayed 10 times within each run in a randomized order.

III. METHOD

A. STFTs

Step 1: convert EEG data to Hjort derivation in order to get the local activity. In this study the EEG data from the c_3 and c_4 electrodes only are analyzed.

Step 2: convert the time domain signal to time-frequency domain signal using STFTs, here we use spectrogram. The EEG data are bandpass filtered and only information between 1 and 45Hz was retained. We use the 73 trials available of the subject k3b's task to test the classification approaches discussed below. Figure 1 shows the STFTs of the c_3 , c_2 and c_4 electrodes of one subject corresponding to left and right hand movement imageries in the selected time window. The STFTs shown in Figure 1 are averaged over several sweeps.



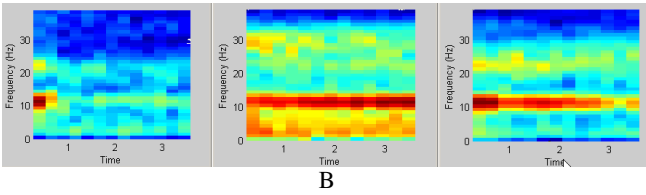


Figure 1. Averaged time–frequency maps of right(R) and left(L) hand movement imagination of subject k3b. A is L and B is R. The figure from the left to right is c_3 , c_2 , c_4 .

Step 3: fisher class separability criterion[14]:

$$F = \frac{(\mu_1 - \mu_2)^2}{\sigma_1^2 + \sigma_2^2} \quad (3)$$

was used to order the features and the first 100 features were selected as input of back-propagation neural networks. In equation (3), μ and σ are the mean and the standard deviation of the feature they correspond to.

Step 4: In order to test our method, we design a linear classifier, and separate the trials into two dataset, one dataset is used to learn, another dataset is used to test, from the three subjects' EEG signals.

B. The SOBI algorithm[15-17]

Step 1: Second-order blind identification (SOBI), a blind source separation (BSS) algorithm was applied to preprocess EEG data. SOBI exploits the time coherence of the source signals to decompose the mixture of sources. SOBI finds W through an iterative process that minimizing the sum-squared cross-correlations between one component at time t and another component at time $t + s$, across a set of time delays. Because such cross-correlations are sensitive to the temporal characteristics within the time series, temporal information contained in the continuous EEG data affects the results of source separation.

Step 2: Same with method 1 step 3.

Step 3: Multilayer back-propagation neural networks were trained to classify motor imagery EEG into two stimulus classes: left and right imagery. The two classes were coded by 1 output unit. The hidden layer consisted of 20 units. The input layer had 200 units, 100 representing channel C3 and 100 for the channel C4. Each input unit represented a feature, selected from the SOBI-recovered SI sources or the EEG channels, covering a time window between 3-7s post-trial onset.

We used the traingdx learning algorithm to train the network. For each of the three subjects, a series of neural networks were trained to 99.99% classification accuracy using 50% of the trials (training set). Subsequently, the trained network was used to classify the remaining 50% of trials (testing set). This two-step process was repeated 60 times with different training sets to obtain the average for the data from each subject.

IV. RESULT AND DISCUSSION

For method 1, we draw some conclusions as following:

One: finding the relation between the fisher distance and ratio. using the same number of feature characteristics, we find the bigger fisher distance, the better classification ratio can be getted. Because alpha (8~17) rhythm' distance of Fisher is the farthest, the character we select are all in alpha (8~17) rhythm^[5].

Two: getting a better classification ratio. The classification performance of the proposed algorithm varied between 70% and 92.6% across subjects.

Three: finding during different time period has different classification ratio. The mainly signals is concentrated on the time period from the third second to the seventh second, in order to select the time period of the true signals, we divided this time period(from the third second to the seventh second)into four time period: one second(the third second to the fourth second), two second (the third second to the fifth second), three second(the third second to the sixth second),and four second(the third second to the seventh second). When we use different time period of three subjects' EEG signals to test our method, we get the relation between the time period and the classification ratio table as table 1.

Table I. The influence of the time period to classification ratio.

	K3b	L1b	K6b	Classification ratio
One second	0.778	0.714	0.650	0.714
two second	0.819	0.786	0.600	0.735
three second	0.848	0.857	0.625	0.777
four second	0.931	0.714	0.650	0.765

For method 2, we draw some conclusions as following.

One: effect to classification accuracy of SOBI algorithm.

We compared classification accuracy using the SOBI-recovered SI components to using the two EEG channels recording. When using the two SOBI-recovered SI sources as inputs, the neural networks trained on 50% of the trials led to a classification accuracy of 88.9% (subject 1) on the remaining 50% of the trials. In contrast, when using the two EEG channels recording, similarly trained neural networks only reached a classification accuracy of 70.2% (subject 1), which was significantly lower than that obtained from the SOBI-recovered left and right SIs (Figure 2). Generalization from the training to testing sets was significantly improved by SOBI preprocessing.

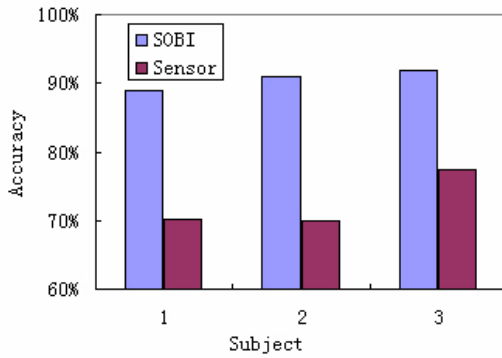


Figure 2. Motor Imagery EEG classification accuracy.

Two: effect to classification accuracy of training sets.

Different classification accuracy has been obtained with different training sets for each subject. The range in classification accuracy across training sets was relatively large in subject 2 (89.25% to 92.25%) and 3 (87.3% to 94.0%) (Figure 3), which means larger error, because of the less trials number of subject 2 and 3 than subject 1.

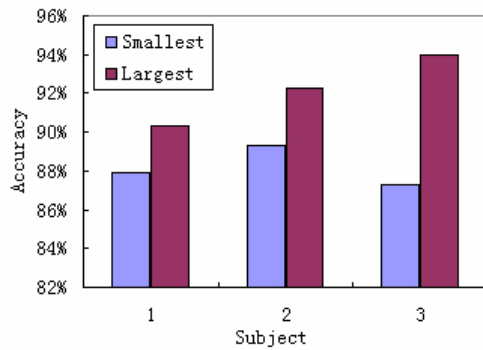


Figure 3. The largest and smallest classification accuracy.

Three: effect to classification accuracy of time windows.

Classification accuracy of different time windows is presented in table 1. For subject 1 and 3, the classification accuracy became larger and larger with the time window became longer and longer. It shows that the features of signal equally distributing on the time of 3-7th second.

Table II. Classification accuracy of different time windows.

Subject	3-4 th second	3-5 th second	3-6 th second	3-7 th second
1	0.651	0.700	0.763	0.889
2	0.860	0.911	0.917	0.909
3	0.683	0.781	0.828	0.899

STFTs and SOBI both are promising algorithm for the analysis of EEG data, by which classification accuracy of Motor Imagery EEG was significantly improved, deserving to be carefully validated and studied for their own sake. We hope to

increase the awareness of the EEG community to this two algorithm and to stimulate future work that compares and contrasts different ways of decomposing EEG data.

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